

Question Paper

Exam Date & Time: 28-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SOFTWARE PROJECT MANAGEMENT [IT_3125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Consider a company managing 2 projects in parallel agendas. One is a software development project - Campus management system, another one is construction project - Building a campus). (5)
- A) The budgetary and timelines are similar ones.
Apply Project Management principles to identify 3 key differentiations among the two projects.
Justify your answer

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A software firm plans to develop a hospital management system with an initial investment of ₹15 lakh. The expected cash inflows are ₹5 lakh, ₹7 lakh, and ₹9 lakh over the next three years. The management wants to decide whether to proceed with the project. Apply cost-benefit evaluation techniques to determine whether this project appears financially viable. Mention the approach or metric you would use and justify your choice. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) A company launched a Digital Transformation Programme that includes projects for cloud migration, cybersecurity enhancement, and customer data analytics. Management needs to ensure that each project contributes to the organization's long-term strategic goals. Apply the concept of Strategic Assessment to explain how the management can evaluate the alignment of each project with the organization's goals mentioning the smart objectives. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) A university IT team plans to develop a student attendance tracking system. Initially, the project scope is unclear, and several departments have conflicting requirements. The project manager decides to follow the Stepwise Project Planning framework to bring clarity and structure. Apply the first three steps of the Stepwise Planning approach to show how the manager can establish project objectives, identify infrastructure, and analyze project characteristics effectively. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Assume that size of an organic type of software product is estimated to be 32,000 lines of source code. Assume that the average salary of a software developer is \$2,000 per month. Determine the effort required to develop the software product, the nominal development time, and the staff cost to develop the product. [a = 2.4, b = 1.05, c = 2.5, d = 0.38] (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) A software company is tasked with developing a banking transaction monitoring system where (2)

requirements are clearly defined and quality assurance at each stage is crucial. Apply your understanding of software development models to identify which model - Waterfall or V-Model - would be more appropriate for this project and justify your choice briefly.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) An IT company is developing a cloud-based customer relationship management (CRM) system comprising modules for lead tracking, sales analytics, and customer support integration. The project involves multiple sequential and parallel activities such as requirement gathering, interface design, database configuration, integration testing, and deployment. (5)
- A) During execution, the manager observes bottlenecks in integration testing and delayed feedback loops from the design team. Apply the Critical Path Method (CPM) to explain how the project manager can determine which activities are critical, calculate the slack time, and revise the schedule to ensure timely project completion. Discuss how CPM aids in prioritizing resource allocation and controlling project delays.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) An IT services firm is executing two concurrent projects - one focused on AI-based customer support automation and another on data analytics dashboard development. Both projects depend heavily on the same machine learning engineer and data analyst, leading to overlapping deadlines and reduced productivity. Analyse the resource allocation challenge presented in this situation and explain how the project manager can apply resource levelling or resource smoothing techniques to optimize workload distribution, minimize schedule conflicts, and ensure consistent project performance without compromising quality. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) A technology firm is developing a cloud-based supply chain management system. The project involves uncertainties related to vendor API integration reliability, data synchronization delays, and fluctuating network performance. Apply the principles of risk management to illustrate how the project team can identify, analyze, and prioritize these risks, and develop appropriate mitigation and contingency plans. Further, explain how Monte Carlo Simulation can be used to quantify the impact of these uncertainties on the project's overall completion time and support better decision-making in schedule and resource planning. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4) An IT organization is executing a Smart City project involving multiple sub-teams working on IoT integration, data analytics, and mobile applications. The project faces coordination challenges, inconsistent reporting, and delays due to unclear responsibilities. (5)
- A) Apply an appropriate Management and Control framework to demonstrate how the project manager can integrate performance metrics, responsibility matrices, and review mechanisms to enhance coordination and control across teams.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) During the development of a banking software system, several developers are working on different modules simultaneously. Multiple versions of source code, documentation, and test cases lead to confusion about which version is the latest stable release. (3)
- Apply the principles of Software Configuration Management (SCM) to describe how the team can maintain version control, ensure consistency among project artifacts, and track changes efficiently throughout the development process.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) A startup hires an external vendor to develop a mobile banking application. Midway through, disputes arise regarding additional features and payment terms due to unclear documentation. (2)
- Explain the importance of a well-defined contract management process in preventing misunderstandings and ensuring smooth coordination between the client and vendor.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5) A mid-size fintech company is expanding its product team to build a new real-time payments gateway. The hiring panel has shortlisted five candidates for the Technical Lead role. The existing team has strong backend and DevOps skills but limited experience in payment security, regulatory compliance, and client stakeholder management. (5)
- A)

Analyse the candidate profiles (assume they vary in technical depth, security/compliance experience, leadership ability, and client-facing skills). Recommend the best candidate for the Technical Lead role and justify your choice considering:

- technical fit for a payments gateway,
- ability to fill the team's current skill gaps (security & compliance),
- leadership and stakeholder management to coordinate with regulators and clients, and long-term team development and organizational goals.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Evaluate how Maslow's Hierarchy of Needs can be used by project managers to improve motivation and performance of software development teams working on long-term projects with high workload and stress levels. (3)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- C) Explain the importance of conducting a post-implementation review during project closure and how it contributes to organizational learning and future project improvement. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_3125:SOFTWARE PROJECT MANAGEMENT

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	20
3	3	CO 3	10
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_3125:SOFTWARE PROJECT MANAGEMENT

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	4(8%)
3	Application	9	33(66%)
4	Analysis	3	10(20%)
5	Create	1	3(6%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)